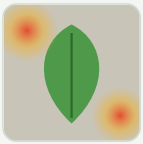
DEEP LEARNING PROJECT · PS5 · EXPLAINABILITY AUDIT

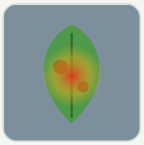
Does the AI look at the leaf, or at the background?

An audit of three plant-disease classifiers using Grad-CAM heatmaps, plus a training fix that forces them to look at the leaf.



Normal training

looks at background



BG-Suppress

looks at the leaf

THE PROBLEM

99% in the lab, poor on the farm

- Plant-disease models score **99%+** on PlantVillage lab photos.
- Those photos have **plain, similar backgrounds** and controlled light.
- So a model can "cheat": learn the background instead of the disease spots. This is called **shortcut learning**.
- On real field photos the same models fail.

PS5 asks: use Grad-CAM to show whether models look at the lesion or the background, and propose a background-suppression training strategy.

HOW I ADDRESSED IT

Audit, prove, fix, re-test

1. **Train 3 models**
   ResNet-50, EfficientNet-B0, ViT-S/16 · 30 epochs each
2. **Grad-CAM → Lesion Focus Score**
   Share of the heatmap that falls on the leaf (0 to 1)
3. **Shortcut tests**
   Leaf removed (background only) and leaf pasted on another class's background (Mixed-Rand)
4. **BG-Suppress training (my fix)**
   Random background swaps + a loss that penalises attention outside the leaf
5. **Field test**
   744 real-farm PlantDoc photos the models never saw

WHAT I FOUND

Shortcut proven, then removed

22–28% accuracy with **no leaf at all** (chance is 10%), so the background does carry information

0.60 → 0.98 ViT leaf focus before → after BG-Suppress (all models end at 0.98–0.99)

13.8 → 0.3 points lost when the background is swapped, before → after

≤ 0.37 points of lab accuracy given up for the fix (still 99.2–99.6%)

25–29% field accuracy for every model; field F1 rose by 1–3 points

Models & tech stack used

ResNet-50

`Residual CNN · 25.6M params`

The classic backbone. Least background-reliant baseline, best field accuracy.

EfficientNet-B0

`Mobile CNN · 5.3M params`

Smallest and phone-friendly. Same lab accuracy with a fifth of the size.

ViT-S/16

`Vision Transformer · 22M params`

Most accurate but most background-reliant. Gained the most from BG-Suppress.

DATA

- PlantVillage tomato · 18,155 images · 10 classes
- PlantDoc · 744 field images
- Leaf masks from segmented copy

FRAMEWORKS

- Python 3.12
- PyTorch 2.11
- timm
- CUDA 12.8

EXPLAINABILITY & ANALYSIS

- Grad-CAM
- scikit-learn
- pandas
- matplotlib

TRAINING SETUP

- RTX 5060 Laptop · 8 GB
- AdamW + cosine LR
- 30 epochs · batch 32
- ImageNet pretrained

ONE-LINE TAKEAWAY

High lab accuracy can hide cheating. Grad-CAM plus background tests **expose the shortcut**, BG-Suppress **removes it**, but real-farm accuracy needs more than a background fix.